



HOME ASSIGNMENT

Welcome to Madbox's **Game Developer** Test 🤘

Goal

🎯 You are to add game feel on top of a pre-existing RPG game. Simple meta systems or features are welcome if they serve that goal, but keep them small — we mainly want this game to "feel" better.

Evaluation

This test helps us evaluate your hard skills in **Game Development**, **Game Feel**, and **performance optimization**. We will look at:

- **Game feel:** what feedback you decided to implement, and how it makes the game more engaging and satisfying.
- **Optimization:** how you keep the frame budget under control. This covers both the cost of the feedback you add (pooling, batching, allocation-free update loops, particle and audio budgets, overdraw) and any optimization work you do on the existing project (spotting and fixing what's already slow).
- **Measurement:** we care as much about *how you know* something is faster as about the fix itself. Profile, measure, then optimize. Tell us the numbers.
- **Code quality:** clean, readable, and something you could hand to a teammate.

When you are done we will review your project and will schedule a review session with you to discuss about it. As a second part of this interview, we will have a small live exercise that will consist in adding a feature to this prototype using AI. This will allow us to evaluate how you can work with AI on a day to day basis.

Do not exceed 8h of work on this project: keep things small, simple, clean. The use of AI tools (e.g. code assistants, generative AI, etc.) is encouraged. However, all code, systems, and design decisions whether created manually or with AI must be fully understood by you and might need to be clearly explained during the review meeting.

Materials

You have received the base project in a zip file.

Delivery

When you are done, please send us your source code (via GitHub or any other online repository) so that we can make a code review.

The repository should also include a **README.md document** in which you'll mention

- **Your approach:** what feedback you decided to add and why.
- **Your optimization work:** what you profiled, what you found, what you changed, and the before/after impact (frame time, GC allocations, draw calls, whatever is relevant). Screenshots of the Profiler are welcome.
- **Trade-offs:** anything you deliberately chose *not* to do for performance reasons, and anything you'd optimize with more time.
- **Time spent** on each thing you implemented.
- **Feedback** on the base project you've been provided.
- Any other comment you may have.

